

FINAL PROJECT REPORT

Multi-Horizon Traffic and Toll Revenue Forecasting for the Padma Bridge

Next-day, weekly and monthly forecasting with horizon-aware models

CSE 4112: Machine Learning Laboratory
Group B2 | Batch 2021
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Ashique portion: multi-horizon forecasting update

Executive summary

This project extends one-day Padma Bridge forecasting to a global direct system covering horizons 1–30. Historical origin dates were expanded into **44,265** origin–horizon examples with **91** predictors. All samples from one origin remained in the same chronological partition, and no observed bridge value after the origin was used.

Across the complete held-out horizon set, XGBoost was the best traffic model with MAE 2,415 vehicles and MAPE 14.21%. CatBoost was the best toll model with MAE BDT 2.580 million and MAPE 14.16%. Performance declines with horizon, confirming the increased uncertainty of longer-range prediction.

Main finding

A single horizon-aware model can learn next-day, weekly and monthly relationships while preserving origin-date integrity. Tree boosting is more reliable than persistence at longer horizons, although rare Eid periods remain substantially harder.

Work completed

- Generated horizon-labelled targets from 1 to 30 days.
- Created origin-safe traffic/toll history and target-date calendar, event and weather inputs.
- Trained and saved Random Forest, XGBoost, CatBoost and Residual MLP models for both targets.
- Calculated horizon-band, holiday, Eid, weekly-total and 30-day-total metrics.

1. Introduction

A one-day forecast supports immediate operations, but staffing, maintenance, revenue planning and event preparation require a longer view. Simply applying a one-day model repeatedly would introduce unknown future lags and compound errors. The implemented global-direct approach instead gives every training sample an explicit forecast horizon and predicts each future date directly from a common forecast origin.

1.1 Objectives

- Preserve next-day performance as a baseline.
- Produce daily total traffic and toll predictions for horizons 1–30.
- Compare ensemble and neural models consistently across horizons.
- Measure daily, weekly-total, monthly-total, holiday and Eid errors.
- Document future-weather uncertainty and avoid origin leakage.

1.2 Forecasting relationship

History available on origin date t + target-date calendar/weather + horizon $h \rightarrow$ bridge total on date $t+h$.

2. Data construction

Item	Value
Raw Padma rows	1,520 daily records
Expanded complete samples	44,265
Forecast horizons	1 through 30 days
Predictors	91
Test origins	224
Test-origin range	2026-01-11 – 2026-08-23

2.1 Origin expansion

For every eligible origin date, 30 supervised examples were created. The predictors describe the same origin history but each row receives a different horizon and target date. Traffic and toll targets are shifted backward by h days so that they refer to the correct future observation.

2.2 Target-date information

Weekday, month, day-of-year, weekend, holiday, Eid proximity and rail-operation status are known from the target date. Historical target-date weather is used in the backtest to represent supplied forecast weather. For actual 8–30 day deployment, unavailable weather should be replaced with a genuine forecast or seasonal climatology and accompanied by a larger uncertainty statement.

3. Leakage prevention and chronological split

Every example is assigned to train/development or test according to its forecast-origin date, not its target date. Thus, the 30 rows originating on one date cannot be split across partitions. Observed bridge-history columns stop at the origin date; later traffic and toll values are targets only.

This grouping rule is essential. Splitting expanded rows randomly would allow nearly identical samples from one origin to appear in both training and test data, producing an unrealistically optimistic score.

3.1 Features

Family	Variables
Origin history	Traffic/toll lags 0–28 and rolling mean/standard deviation
Target calendar	Cyclic weekday, month and annual encodings; weekend
Events	Public holiday, Eid and distance to nearest Eid
External	Target weather and railway-operation indicator
Horizon	Integer forecast_horizon from 1 to 30

3.2 Models

Model	Role
Persistence	Latest observed level for all future dates
Seasonal baseline	Most recent known value matching target weekday
Random Forest	Nonlinear bagged trees
XGBoost	Regularized gradient boosting
CatBoost	Alternative boosting model
Residual MLP	Predicts change relative to the origin value

4. Overall forecasting results

Target	Best model	MAE	RMSE	MAPE	R-squared
Toll	CatBoost	2,580,294	4,479,452	14.16%	0.314
Traffic	XGBoost	2,415	4,096	14.21%	0.501

Interpretation

XGBoost explains about half of the variation in 1–30 day traffic targets and provides the lowest traffic MAE. CatBoost provides the lowest toll MAE, while overall toll R-squared is lower at 0.314. This is expected because toll revenue combines traffic variation with changes in vehicle composition and revenue per vehicle, neither of which is directly forecast at long horizons.

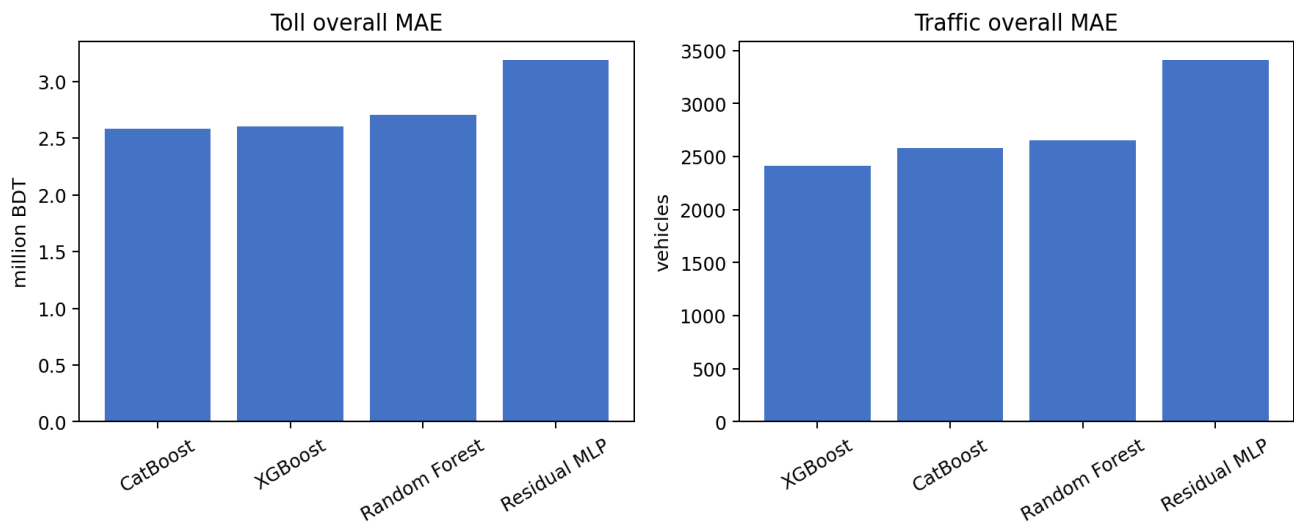


Figure 1. Overall held-out MAE across the four trained model families.

5. Accuracy by forecast horizon

The horizon bands separate immediate prediction (day 1), short-range planning (days 2–7) and longer-range planning (days 8–30). Errors increase because recent observations become less informative and future events/weather are less certain.

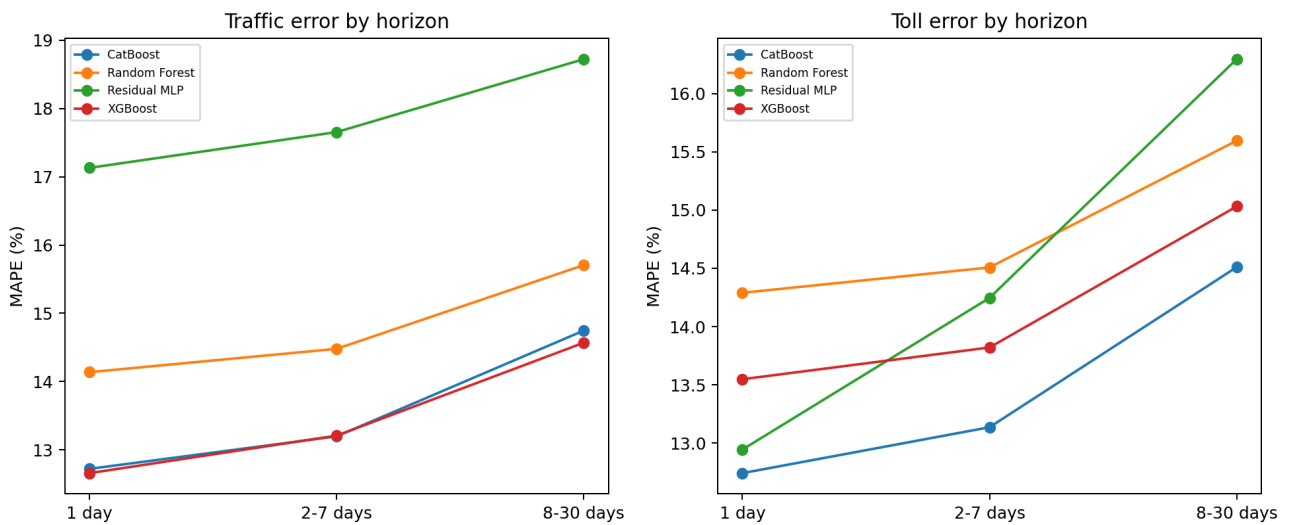


Figure 2. Traffic and toll MAPE by horizon band for every trained model.

Target	Band	Best model	MAE	MAPE	R ²
toll	1 day	CatBoost	2,317,154	12.74%	0.334
toll	2-7 days	CatBoost	2,373,043	13.14%	0.347
toll	8-30 days	CatBoost	2,650,630	14.51%	0.305
traffic	1 day	XGBoost	2,171	12.66%	0.525
traffic	2-7 days	XGBoost	2,239	13.21%	0.516
traffic	8-30 days	XGBoost	2,475	14.57%	0.497

6. Daily prediction comparisons

The plots below show representative test predictions at horizons 1, 7 and 30 for the selected overall model. Day-one estimates respond most closely to recent bridge conditions. At horizon 30, predictions are smoother and extreme peaks are harder to reproduce because the origin contains no intervening observed traffic.

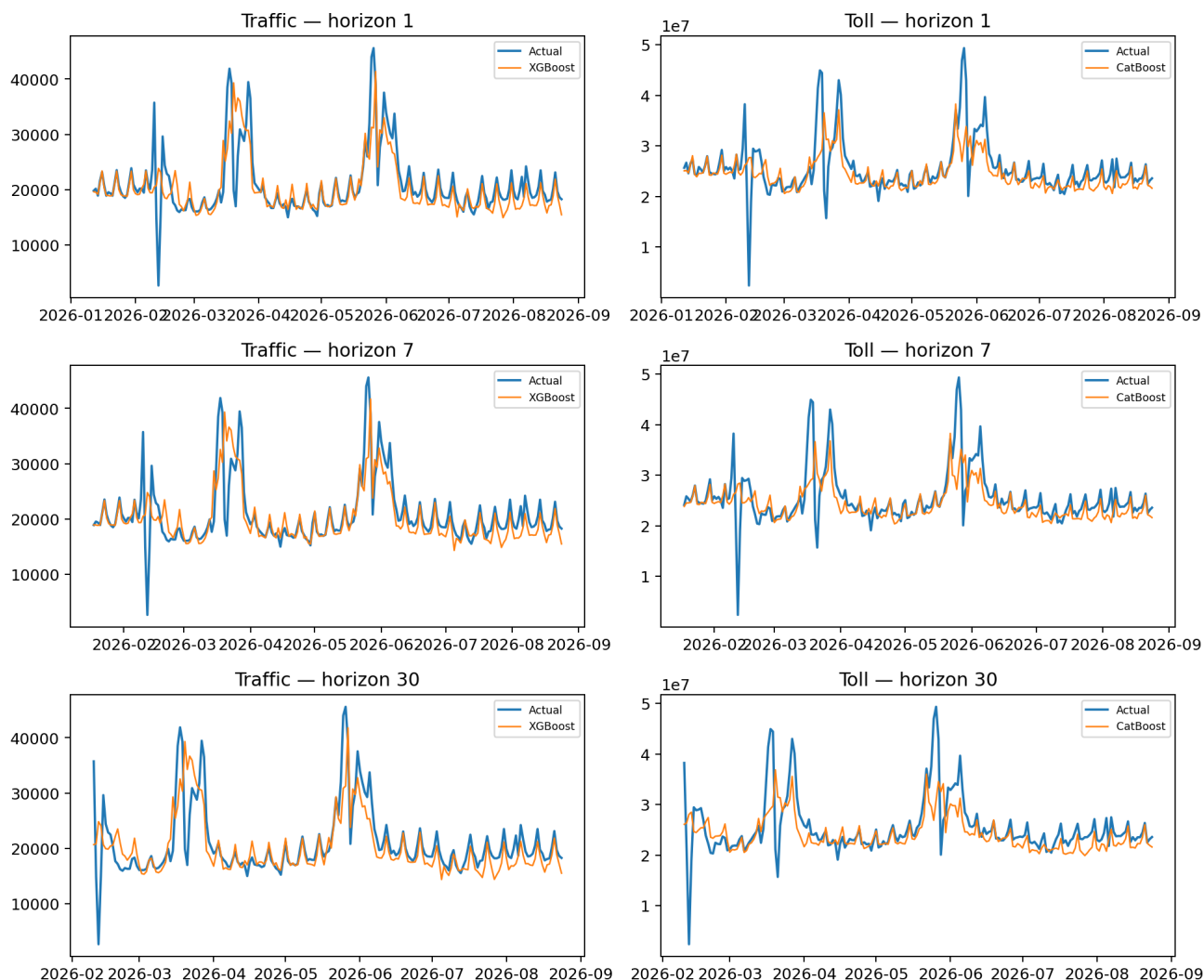


Figure 3. Actual and selected-model forecasts at horizons 1, 7 and 30.

7. Weekly and 30-day aggregate performance

Daily predictions were summed within each forecast origin and compared with the corresponding observed 7-day or 30-day total. Aggregation can cancel some positive and negative daily errors, but systematic bias remains visible.

Target	Model	Period	Aggregate MAE	MAPE	R ²
toll	CatBoost	7 days	10,204,152	5.44%	0.774
toll	XGBoost	7 days	11,030,636	6.00%	0.709
toll	Random Forest	7 days	11,705,324	6.39%	0.610
toll	Residual MLP	7 days	13,257,397	6.83%	0.501
toll	XGBoost	30 days	34,173,632	4.86%	0.941
toll	CatBoost	30 days	37,519,718	5.11%	0.926
toll	Random Forest	30 days	40,650,919	5.90%	0.920
toll	Residual MLP	30 days	47,359,546	6.28%	0.863
traffic	XGBoost	7 days	9,407	6.26%	0.848
traffic	CatBoost	7 days	10,713	6.88%	0.801
traffic	Random Forest	7 days	10,947	7.44%	0.793
traffic	Residual MLP	7 days	15,390	10.80%	0.599
traffic	XGBoost	30 days	32,094	5.66%	0.931
traffic	Random Forest	30 days	35,874	6.54%	0.919
traffic	CatBoost	30 days	41,042	6.71%	0.883
traffic	Residual MLP	30 days	55,158	9.09%	0.762

Weekly totals are suited to staff and lane planning, while 30-day totals can support budgeting. They should not be interpreted as 30 independent one-day forecasts: all predictions in a row share the same origin information.

8. Holiday and Eid performance

Target	Segment	Model	N	MAE	MAPE	R ²
toll	Eid +/-7d	XGBoost	900	7,148,264	24.65%	-0.291
toll	Eid +/-7d	CatBoost	900	7,337,380	23.06%	-0.251
toll	Eid +/-7d	Random Forest	900	7,516,654	27.25%	-0.625
toll	Eid +/-7d	Residual MLP	900	9,483,113	27.69%	-0.835
toll	Eid +/-7d	Persistence	900	10,759,624	30.81%	-1.476
toll	Eid +/-7d	Seasonal-7	900	11,285,177	31.34%	-1.637
toll	holiday	XGBoost	683	8,853,726	77.18%	-0.091
toll	holiday	Random Forest	683	9,031,631	79.54%	-0.243
toll	holiday	CatBoost	683	9,126,664	74.68%	-0.057
toll	holiday	Residual MLP	683	9,202,320	69.75%	-0.109
toll	holiday	Seasonal-7	683	9,990,267	70.18%	-0.384
toll	holiday	Persistence	683	10,339,527	72.87%	-0.429
traffic	Eid +/-7d	XGBoost	900	6,156	21.74%	-0.122
traffic	Eid +/-7d	Random Forest	900	6,429	22.97%	-0.248
traffic	Eid +/-7d	CatBoost	900	7,180	22.95%	-0.415
traffic	Eid +/-7d	Residual MLP	900	8,423	25.88%	-0.830
traffic	Eid +/-7d	Persistence	900	11,818	35.67%	-2.459
traffic	Eid +/-7d	Seasonal-7	900	12,437	36.68%	-2.821
traffic	holiday	XGBoost	683	7,807	62.11%	0.120
traffic	holiday	Random Forest	683	8,457	64.47%	0.004
traffic	holiday	CatBoost	683	8,472	60.79%	0.009
traffic	holiday	Residual MLP	683	9,296	61.59%	-0.128
traffic	holiday	Seasonal-7	683	11,222	62.81%	-0.803
traffic	holiday	Persistence	683	11,399	64.52%	-0.735

Eid errors are much higher than overall errors. The test set contains relatively few distinct Eid events even though each event creates many origin–horizon rows. Those rows are correlated and should not be mistaken for thousands of independent festivals. Event-specific modeling and uncertainty intervals are therefore important next steps.

9. Prediction uncertainty and operational use

The current deliverable produces point predictions. Longer horizons require wider uncertainty bands because weather, traffic composition and unexpected disruptions are not fully known. A future version should add quantile boosting or split-conformal intervals evaluated separately by horizon.

Mode	Recommended interpretation
Next day	Operational traffic/toll estimate using fresh origin history
Next 7 days	Daily schedule plus weekly total; use forecast weather
Next 30 days	Planning scenario; substitute climatology where weather is unavailable
Eid period	High-uncertainty scenario requiring event monitoring

10. Limitations

- Approximately four years of history limit the number of repeated annual events.
- Historical observed weather is more accurate than real long-range weather forecasts.
- Future Jamuna observations are intentionally excluded after the origin.
- Accidents, closures, strikes and toll policy changes are not modeled.
- The compact local training configurations are reproducible but not an exhaustive hyperparameter search.
- Point forecasts do not yet include calibrated prediction intervals.

11. Conclusion

The multi-horizon extension successfully trains a single horizon-aware forecasting system for daily traffic and toll revenue from one to thirty days ahead. XGBoost is the strongest overall traffic model and CatBoost is the strongest overall toll model on test MAE. The results demonstrate useful planning ability while quantifying the expected loss of accuracy as the horizon grows. The most important deployment requirements are origin-safe feature updates, realistic future-weather inputs and explicit uncertainty for Eid and 30-day forecasts.

Appendix A. Delivered artifacts

Eight serialized models, 6,255 row-level test predictions, 72 horizon/event metric rows, weekly and monthly aggregate tables, figures, a run summary and the complete training script are included.

Appendix B. Reproduction

From the multi-horizon project folder, run: `python train_multihorizon.py`. It regenerates the expanded samples, saved models, tables and figures.

References

1. Bangladesh Bridge Authority, Padma Bridge traffic and toll records, <https://bba.gov.bd/>
2. Open-Meteo historical weather documentation, <https://open-meteo.com/en/docs/historical-weather-api>
3. CSE 4112 Group B2 multi-horizon forecasting brief, September 2026.